BY ORDER OF THE COMMANDER
RAF LAKENHEATH (USAFE)

LAKENHEATH INSTRUCTION 15-101

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Weather

WEATHER SUPPORT

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This instruction implements Air Force Policy Directive (AFPD) 15-1, *Weather Operations.* It establishes responsibilities and weather support procedures for Royal Air Force (RAF) Lakenheath and RAF Feltwell. It applies to all units assigned to the 48th Fighter Wing. It provides general information for observations and forecasts; weather warnings, watches, and advisories; space weather supported services and dissemination of information and reciprocal support. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using AF Form 847, *Recommendation for Change of Publication*; route AF Form 847s from the field through the appropriate functional chain of command. This publication may not be further implemented/extended. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS).

SUMMARY OF CHANGES

Resource protection procedures have been updated to reflect up-to-date requirements. Specialized Support procedures have been updated to remove redundant information already available in other authoritative documents. Additional clarification is added for support expectations and requirements from 48 OSS/OSW and other agencies. Outdated information and procedures have been removed. This includes the transition from AFW-WEBS to BIFROST.

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1. General Information.

1.1. Introduction. 48th Operations Support Squadron Weather Flight (48 OSS/OSW) provides and arranges weather support for the 48th Fighter Wing (48 FW). Basic concepts and procedures are outlined in Air Force and Major Command (MAJCOM) directives listed in **Attachment 1**. This instruction establishes responsibilities and procedures for areas of weather support required for 48 FW operations.

1.2. Support. Weather support provided by 48 OSS/OSW is largely driven by the requirements of flying operations and base agencies.

1.3. Duty Priorities. Duty priorities are listed in the RAF Lakenheath Weather Support Operating Procedures maintained by the 48 OSS/OSW.

1.4. Release of Weather Information. Weather observations, forecasts, and historical data/records will be released only to Department of Defense (DoD) agencies or contractors under contract to the United States Air Force (USAF). All other agencies and civilians will be referred to Public Affairs (48 FW/PA) and/or the Base Freedom of Information Act Monitor (48 CS/SCXK) to obtain approval prior to the release of weather observation records.

2. Limitations to Weather Services.

2.1. Interruption of the normal receipt of alphanumeric and graphic data via Joint Environmental Toolkit (JET), Non-classified Internet Protocol Router Network (NIPRNET) or Secret Internet Protocol Router Network (SIPRNET) can degrade forecasting capabilities.

2.2. 48 OSS/OSW does not have weather radar. To compensate, the weather flight uses a combination of Air Traffic Control (ATC) radar, data from the Air Force Weather Web Services (AFW-WEBS), United Kingdom (UK) composite radar and/or pilot reports (PIREPs) to determine if precipitation and/or thunderstorms are within 25 Nautical Miles (NM) of the base.

2.3. RAF Feltwell has no weather personnel or weather sensors. Therefore, in the interest of safety and resource protection, all weather watches, warnings, and advisories (WWAs) issued for RAF Lakenheath are also in effect for RAF Feltwell. The only exceptions are that induction icing and wind shear are not applicable to RAF Feltwell.

2.4. Due to local terrain features near the runway, sensor wind speeds between the 05 and 23 approach ends of the runway may differ by several knots. This is a standard and acceptable variance, and no corrective action is warranted.

2.5. During calendar year 2025, the Joint Environmental Toolkit (JET) system for issuing/amending WWAs will be replaced with BiFROST. Users may experience some technical differences as the transition occurs.

2.6. Communication Outages. 48 OSS/OSW is primarily responsible for all METWATCH, TAF, and resource protection responsibility at RAF Lakenheath and RAF Feltwell. If a communication outage occurs, 48 OSS/OSW will exhaust are resources and notify local agencies of potential weather threats. The 100 OSS/OSW, 420 ABS/OSW, and 557 WW are available to assist 48 OSS/OSW with disseminating TAFs, WWAs, & DD175-1s until communication links are restored.

3. Airfield Support Function (ASF).

3.1. General. All official weather observations at RAF Lakenheath will be taken and augmented IAW AFMAN 15-111, *Surface Weather Observations*. 48 OSS/OSW operates a FMQ-19, automated observing system, which provides 24 hour observation capability. 48 OSS/OSW will be on duty one hour prior to airfield opening until airfield closure, as stated in published FLIP hours. Deviations to these hours can be made by the Weather Flight Commander as required to support mission needs. A standby forecaster will be subject to recall at all times when the weather station is closed. A standby technician may be recalled through the Command Post (CP) at DSN 314-226-4800 or the assigned mobile phone and will report to the weather station no more than one hour after being recalled.

3.2. If the primary observation site, Building 1392, is non-operational, or the technician has to evacuate, observations will be taken from the Alternate Operating Location (AOL). Building 1319 is the primary AOL. However, if Building 1319 is unsafe or inaccessible, any location with a view of the airfield complex, a telephone, and access to the base local area network (LAN) may be used as an AOL in an emergency. Observations made from an AOL might contain estimated elements, such as wind and pressure. Contact information for the AOL is in the RAF Lakenheath Weather Installation Data Page.

3.3. Augmenting Observations. Due to the nature of weather occurring here at RAF Lakenheath, observations may be augmented during poor weather conditions. This is to back-up automated weather sensors that may show erroneous values for visibility and/or ceilings at either end of the runway so that the accuracy of observed conditions increases during poor weather conditions. Additionally, if any conditions listed in AFMAN 15-111, Table 5.1 are observed or forecasted, established backup procedures will be implemented.

4. Mission Integration Function (MIF).

4.1. General. The 48 OSS/OSW is primarily responsible for a Mission Weather Product (MWP) or DD Form 175-1, *Flight Weather Briefing*.

4.2. Local MWP. A local MWP consists of two parts: the Mission Execution Forecast (MEF), a graphical planning document produced daily to support local flying; and a verbal brief provided to each aircrew or squadron Operations Supervisor based on individual mission weather thresholds. These two parts in conjunction are the equivalent of a DD Form 175-1, prescribed by DAFMAN 15-129, *Air and Space Weather Operations*.

4.2.1. A MEF will be disseminated four hours prior to first takeoff unless otherwise specifically requested for an early time 24 hours in advance, via the 48 OSS/OSW SharePoint page. This product provides a mission outlook tailored towards scheduled and common airspaces and ranges for the entire daily flying window for 48 FW operations. It provides an outlook for alternate and divert airfields and winds and wave forecasts. An updated MEF will be posted every eight hours until the end of the flying window. Due to the vastness of the forecast area, the MEF is not tailored for individual missions and thus utilized for planning purposes. An in-place forecaster (manning limitations allowed) will be in each squadron to verbally tailor the MEF for individual mission planning.

Attachment 2 is an example of the MEF. The MEF will be reviewed at a minimum annually with all Fighter Squadron DOs for currency and required mission changes and documented for OSW records. The 48 OSS/OSW Mission Integration or Staff Integration NCOICs will coordinate changes with the 48 OSS, 495 FS, 494 FS, 493 FS and 492 FS Directors of Operations before changes to content or format are made.

4.2.2. Briefing Procedures. Mission weather briefers will be present at each fighter squadron at least two hours prior to each subsequent takeoff until final land to assist with mission planning as manning limitations allow. If an in-person briefer is unavailable due to manning restraints, a Flight Weather Brief can be obtained by calling 314-226-4184. A briefer will provide a Flight Weather Brief via PowerPoint presentation and verbal brief during the aircrew step brief, which is normally one hour prior to takeoff. This PowerPoint presentation includes all information required by DAFMAN 15-129, *Air and Space Weather Operations*, and will be retained by 48 OSS/OSW in accordance with AF Records Disposition Schedule and AFMAN 33-322, *Records Management and Information Governance Program.*

4.2.3. Updates to MWPs. Updates to each MWP will be communicated to aircrew or squadron Operations Supervisors prior to aircrew step brief time.

4.2.4. Amendments. The MWP will be amended IAW DAFMAN 15-129, Chapter 4, and any time it is deemed unrepresentative of the actual or forecast conditions.

4.3. Planning Weather. A planning forecast for the next flying day will also be completed each duty day and posted on the 48 OSS/OSW SharePoint page. The format will be similar to the Mission Weather Product. The first planning forecast will be posted simultaneously with the first MEF and will only provide a winds and waves forecast. The second planning weather will be posted by 1300L and will include a regional weather outlook tailored towards scheduled and common airspaces and ranges and an update to winds and waves. The planning weather will not be produced prior to a 3-4 day weekend due to weather model reliability.

4.4. MISSIONWATCH. A MISSIONWATCH is conducted by the MIF forecaster on all areas of operations and will use risk management to focus on daily mission areas to be used by the 48 FW, IAW DAFMAN 15-129, Chapter 4. If the MIF technician determines the need to notify aircrew of weather changes discovered during MISSIONWATCH, they will immediately contact the Supervisor of Flying (SOF) and the flying squadron Operations Supervisors and provide pertinent information to these individuals for relay to aircrew.

4.5. Pilot-to-Metro Service (PMSV). 48 OSS/OSW provides PMSV support as listed in the Flight Information Publication (FLIP). Weather operates this program IAW DAFMAN 15-129, Chapter 5, and uses standard phraseology found in Federal Aviation Administration Handbook (FAAH) JO 7110.10, Chapter 12.

4.6. Electro-Optical (EO) Support. Lakenheath uses the Target Acquisition Weapons Software (TAWS) to generate electro-optical tactical decision aid (EOTDA) products. This windows-based program computes detection and lock-on ranges for various precision-guided munitions (PGMs) and temperature contrasts between target and background. Aircrew will submit request for TAWS data by filling out a request form on the 48 FW Weather SharePoint page, or by calling 226-4184. TAWS products should be requested at least 2 hours prior to the desired delivery time. Short notice requests will be handled by the counter forecaster as duty priorities allow.

5. 557 Weather Wing (557 WW).

5.1. General. The 48 OSS/OSW works in concert with the 557 WW and 14 Weather squadron (14 WS). The 557 WW is open 24 hours per day, 7 days a week and is located at Offutt Air Force Base, Nebraska. The 14 WS serves as the climatology center for Air Force weather. The 100 OSS/OSW and 420 ABS/OSW are alternate backups for all Terminal Aerodrome Forecasts (TAFs), forecast watches and observed advisories (when required) when the 48 OSS/OSW is incapable of issuing these products during major communication outages. If unable to support, 48 OSS/OSW will reach out to USAFE/A3W and the 577 WW to input a Special Assistance Request (SAR).

6. Specialized Support.

6.1. General. Unless stated otherwise in this instruction, the minimum notification time required to request briefing support is 24 hours. Short-notice requests will be fulfilled if weather personnel are available. 48 OSS/OSW standard briefing format is PowerPoint and/or paper hardcopies.

6.2. Transient Aircraft. 48 OSS/OSW will provide transient aircraft weather briefings via DD175-1s upon request when a technician is on duty and time allows. When no MIF technician is on duty, or when higher duty priorities preclude it, the 557 WW will provide the flight weather briefings. The 557 WW can be contacted at DSN 489-2133/6145, via email at **ows.ops@us.af.mil.**

6.3. Cross-Country. 48 FW aircrews flying outside of the MEF area will contact 48 OSS/OSW to obtain a **DD175-1**. The minimum notification time for a DD175-1s is two hours. 48 OSS/OSW will provide support to all 48 FW aircraft while off-station via DD Form 175-1 or MEF support tailored to the mission area via in-squadron briefings or reach-back support from RAF Lakenheath. When 48 FW are moving to/from CONUS or out of the EUCOM theater, the briefing typically will be conducted via the ACC Air Operations Squadron, Air Operations Weather (AOS/AOW) IAW DAFMAN15-129, Table 4.3**,** and briefed by 48 OSS/OSW mission forecasters. There will be no DD175-1s created by 48 OSS/OSW in this situation. The minimum notification time for a new MEF location request is 1 week.

6.4. Deployment/Contingency. 48 OSS/OSW will provide in-squadron briefings in support of deployments and/or contingency operations. General information required for briefing support is briefing time and location, take-off times, route, destination, and CORONET Delivery Control Officer (DCO) contact information if available.

6.5. Situation Briefings. Situation briefings are briefed upon request. Content varies depending upon the situation; however, most briefings contain a satellite snapshot and the local forecast.

6.6. CORONET. These briefings are prepared by the AOS/AOW. 48 OSS/OSW can assist with a weather presentation upon request from the coordinating Air Combat Command Delivery Control Officer (ACC DCO), who will establish brief times and locations, but it will not differ from the AOS/AOW forecast.

6.7. SAAM Support. 48 OSS/OSW will provide weather support when RAF Lakenheath is conducting close watch exercises or operations.

6.8. Instrument Refresher Course (IRC). A climatological presentation will be briefed by 48 OSS/OSW when requested by instructor. This briefing will include information seasonal weather hazards and climatological trends to assist mandatory continuation training for aircrew.

6.9. Base/School delays and cancellations. When requested, 48 OSS/OSW will provide weather information to the Mission Support Group Commander (48 MSG/CC) for impacts to base agencies.

6.10. Chemical Downwind Messages (CDM) and Effective Downwind Messages (EDM). 48 OSS/OSW will provide weather information for creation of CDMs and EDMs as required. Both can be pulled immediately utilizing the CDM and EDM tab on C2IMERA or upon request.

6.11. Long-range forecast. 48 OSS/OSW creates a 5-day weather outlook by 0830L each duty day for RAF Lakenheath and forward operating locations containing 48 FW aircraft. A 7-day forecast will be provided prior to 4-day weekends. This will be posted to 48 OSS/OSW SharePoint, AF Connect App, and C2IMERA.

6.12. Tropical Cyclone Support. 48 OSS/OSW will not deviate from official National Hurricane Center and Joint Typhoon Warning Center forecasts on tropical storm tracks or wind forecasts but does have leeway to depict localized weather effects expected at RAF Lakenheath and RAF Feltwell. 48 OSS/OSW will provide information to Emergency Management and 48 FW Commanders to necessary to make mission essential decisions such as evacuation and resource protection. Forecasts will be made a minimum of 96-hours prior to tropical storm impact.

6.13. Volcano Support. 48 OSS/OSW will use the appropriate Volcanic Ash Advisory Center (VAAC) to provide necessary forecast services/products required for 48 FW Commanders to necessary to make mission essential decisions such as evacuation and resource protection.

6.14. Climatological Support. Requests for climatological data and studies should be sent to 48 OSS/OSW Flight Leadership. Minimum request time is four duty days.

7. Weather Watches, Warnings, and Advisories (WWA).

7.1. General. Weather WWAs will be issued when any of the specific criteria listed in **Attachment 3** is occurring or is expected to occur, as appropriate, within a 5 NM radius (unless otherwise specified) from the center of the RAF Lakenheath runway. In the interest of safety and resource protection, all WWAs issued for RAF Lakenheath are also in effect for RAF Feltwell with the exceptions of induction icing and wind shear. 48 OSS/OSW is the primary issuing authority for WWAs. When 48 OSS/OSW is unable to do so and there is imminent risk to life and/or property, then by 100 OSS/OSW or 420 ABS/OSW will issue WWAs in accordance with the established COOP MOA.

7.2. Definitions:

7.2.1. Weather Advisory (WA). A special notice provided to supported customers that alerts them when an established weather condition that could affect operations is occurring. Advisory criteria are located in **Attachment 3.**

7.2.2. Weather Watch (WWATCH). A special notice provided to supported customers that alerts them of conditions favorable for the development (potential) of weather conditions within 5 NM of the center point of the RAF Lakenheath runway of such intensity as to pose a hazard to life or property. Watch criteria are located in **Attachment 3**.

7.2.3. Weather Warning (WW). A special notice provided to supported customers that alerts them to the occurrence or expected occurrence of weather conditions within 5 NM of the center point of the runway of such intensity as to pose a hazard to life or property. Warning criteria are located in **Attachment 3**.

7.3. Dissemination.

7.3.1. Issuing and Canceling. Weather WWA meeting the criteria on the **Attachment 3** will be disseminated IAW Attachments **4 and Attachment 5**, with BiFROST being the primary means.

7.3.2. Observed WWA. Observed advisories will be issued by the 48 OSS/OSW during airfield hours of operations. During hours of closure, all observed advisory issuance will be automated, with the exception of observed lightning within 5 nautical miles of RAF Lakenheath and Feltwell when the standby forecaster will issue manually.

7.3.3. Forecast Warning. Severe and non-severe forecast warnings will be issued by the 48 OSS/OSW. During hours of closure, the standby forecaster will report NLT 1 hour prior to issuance.

7.3.4. Forecast Watch. Severe and non-severe forecast watches, will be issued by the 48 OSS/OSW. During hours of closure, the standby forecaster will report NLT 1 hour prior to issuance.

7.3.5. Confirmation of Receipt. The weather flight will verify receipt by the SOF, CP, RAPCON, Airfield Management Operations (AMOPS), Tower, and MOC.

7.3.6. Backup Procedures. If there is a BiFROST outage, then telephones and hotlines will be used to disseminate weather WWAs to normal recipients, and to notify them of the outage.

7.4. Severe Weather Action Plan (SWAP). When severe weather threatens RAF Lakenheath and/or RAF Feltwell, 48 OSS/OSW will take actions to monitor the weather and provide timely notification to base agencies. Standby personnel will also be available for recall at all times and will man the weather station IAW section 3.1 of this document. Once recalled (or notified of the need during regular duty hours), 48 OSS/OSW will augment personnel as the situation dictates and intensify the METWATCH of all weather changes and notify impacts to base agencies, within duty priorities. The weather flight will identify in local standard operating procedures (SOPs) further SWAP actions required (e.g. checklists accomplished). In addition, supported agencies can request that the 48 OSS/OSW take specific actions for each type of severe weather.

8. Roles and Responsibilities.

8.1. In order for the 48 OSS/OSW to provide the best weather support possible it is necessary to receive reciprocal support from various organizations.

8.2. 48 Fighter Wing Command Post (48 FW/CP) will:

8.2.1. Upon notification from 48 OSS/OSW of WWA issuance for RAF Lakenheath and RAF Feltwell immediately notify all base agencies IAW Attachments **4 and Attachment 5**. When the airfield is closed, the standby forecaster can be recalled fulfilling operational or resource protection weather requirements as needed.

8.2.2. Ensure the 48 OSS/OSW is promptly notified of all contingencies, alerts, exercises, deployments, or emergency movement of aircraft that would require weather support.

8.2.3. Promptly notify 48 OSS/OSW of any aircraft incidents or mishaps or any weather-related property damage to the base, so they can save significant weather data.

8.2.4. Notify the 48 OSS/OSW on all OPREP-3 messages on events or incidents that involve weather to ensure accurate weather data and information is contained in the OPREP-3 message.

8.3. 48 FW Plans and Inspections (48 FW/XP) will reserve workspace for one weather person, including one NIPRNET and one SIPRNET connection, and a secure telephone in the Installation Control Center (ICC) for exercises and contingency operations.

8.4. 48 FW Inspector General (48 FW/IG) will ensure 48 OSS/OSW conducts Severe Weather Action Procedure (SWAP) and Alternate Operating Location (AOL) exercises annually.

8.5. 48 FW Safety Office (48 FW/SE) will notify 48 OSS/OSW as soon as practical of all reported air or ground incidents in which weather or weather services may be a factor.

8.6. 48 FW Installation Deployment Officer (48 LRS/LGRDX) will ensure a weather briefing (if requested) is included in the concept briefing, to include the appropriate deployment area. 48 OSS/OSW requests as much notice as possible for concept weather briefings.

8.7. 48th Communications Squadron (48 CS) will coordinate local and enterprise support to mitigate and resolve outages to 48 FW systems that are used to obtain weather data. Support priorities will be triaged based on 48 FW's mission priorities. Critical weather systems, Joint Environmental Toolkit server, Airfield Services and Mission Integration computer systems are critical to safety of flight and should be prioritized to be fixed as soon as possible.

8.7.1. The following order of priority will apply to restoring communications:

8.7.1.1. SIPRNET.

8.7.1.2. NIPRNET

8.7.1.3. PMSV Radio.

8.7.1.4. Phone.

8.7.2. Provide all position certified 48 OSS/OSW personnel with permission to upload documents within the shared documents section of the weather pages on the 48 FW Weather SharePoint page.

8.7.3. Provide communications and equipment plans and programming support on matters affecting RAF Lakenheath weather equipment.

8.8. 48th Civil Engineering Squadron Installation Office of Emergency Management (48 CES/CEX) will:

8.8.1. Ensure the Installation Emergency Management Plan (IEMP) 10-2 and procedures that require weather support are coordinated with 48 OSS/OSW.

8.8.2. Ensure the 48 OSS/OSW Flight Commander or designated alternate attends the semi-annual in-person Emergency Management Working Group to address any issues regarding severe weather resource protection.

8.8.3. Solicit required weather data from 48 OSS/OSW for CDM and EDM.

8.9. Defense Infrastructure Organization (48 CES/DIO) will obtain permission from the on-duty weather shift supervisor prior to transferring any weather facility or weather sensor equipment to/from primary power.

8.10. 48th Operations Group Standardization and Evaluation (48 OG/OGV) will ensure that SOF interaction with 48 OSS/OSW will be conducted IAW AFI 11-418 Lakenheath Supplement, *Operations Supervision*.

8.11. Flying Squadron Responsibilities:

8.11.1. Aircrew desiring mass weather briefings (in squadrons, etc.) will request these briefings at least 24 hours in advance. Short-notice requests will be fulfilled upon weather personnel availability and 48 FW priorities.

8.11.2. Aircrew flying local sorties will provide post-mission weather feedback to 48 OSS/OSW whenever possible, especially if conditions were significantly different than the forecast or could significantly impact future flights.

8.11.3. Coordinate weather support requirements at least 48 hours in advance when possible, with 48 OSS/OSW Flight Commander or designated alternate for all deployments. Requirements should include both support for flying operations and resource protection.

8.11.4. Provide a workspace with NIPR capability and opportunity for at least one weather technician to perform in-squadron MWP refinement and aircrew pre-departure briefings at or near the operations counter, both home station and deployed. Provide the code for entry into Buildings 1319 and 1355 to the Weather Flight Commander or Flight Chief any time the codes are changed, to include cipher locks on the outside doors.

8.11.5. Report PIREPs as soon as possible after making the observation. PIREPs at any time are valuable, especially in poor flying weather. In-flight PIREPs may be passed to any USAFE Control Agency (Control Tower, Ground Radar Facility, etc.) or RAF Lakenheath PMSV. Post-flight PIREPs may be passed directly to weather personnel or by phone. PIREPs passed to any USAFE Air Traffic Control Agency at RAF Lakenheath will be relayed to the ASF technician as soon as possible. When relaying a PIREP, aircrew will attempt to report the following information:

8.11.5.1. Time, location, altitude, and aircraft type.

8.11.5.2. Clouds (bases, amount (FEW, SCT, BKN, OVC), tops) on takeoff, enroute, and arrival.

8.11.5.3. Hazards: icing, turbulence, hail, lightning, etc. Be sure to specify intensity, location, and proximity to clouds, altitude, and time.

8.11.5.4. Significant deviations from forecasted weather, winds, clouds, etc.

8.11.5.5. Wind shear whenever encountered.

8.12. 48th Operations Support Squadron Intel Flight (48 OSS/IN) will:

8.12.1. Ensure 48 OSS/OSW is informed when a contingency mission-planning cell is activated.

8.12.2. Notify 48 OSS/OSW as soon as possible when requesting Mission Qualification Training briefing support.

8.12.3. Coordinate weather support requirements as dictated by 48 FW operations and exercises. Give at least 48 hours to collect data and build products.

8.13. 48th Operations Support Squadron Current Operations Flight (48 OSS/OSO) will:

8.13.1. Ensure 48 OSS/OSW has access to weekly/daily flying schedule and other information required for flight weather briefings.

8.13.2. Promptly notify 48 OSS/OSW of upcoming IRC briefings.

8.14. 48th Operations Support Squadron Airfield Operations Flight (48 OSS/OSA) will:

8.14.1. Notify 48 OSS/OSW leadership of quarterly Airfield Operations Board meetings which 48 OSS/OSW are required to attend.

8.14.2. Submit change requests to FLIP on behalf of 48 OSS/OSW. If the change is urgent, the NCOIC, Airfield Management Operations (NAMO), or designated FLIP manager will issue a Notice to Airmen (NOTAM).

8.14.3. Cooperative Weather Watch (CWW). ATC units play an essential role in the CWW by reporting the following data in a timely manner to 48 OSS/OSW:

8.14.3.1. Pilot Reports (PIREPs). RAPCON and Tower will:

8.14.3.1.1. Solicit PIREPs from aircraft IAW JO 7110.65, *Air Traffic Control*, and/or when requested by 48 OSS/OSW personnel.

8.14.3.1.2. The ATC facility that receives a PIREP will relay the information to Weather Flight personnel as soon as practical, within ATC established duty priorities, preferably within 15 minutes due to the nature of weather data.

8.14.3.2. IAW JO 7110.65, and AFI 13-204V3, ATC duties take precedence over non-ATC duties. Controllers shall not allow the reporting of meteorological information to interfere with air traffic control operations.

8.14.3.3. All tower controllers must receive CWW training IAW DAFMAN 13-204V3 and AFMAN 15-111.

8.14.3.4. 48 OSS/OSW will assist tower personnel in creating appropriate guides to perform CWW. This will include day and night visibility marker guides that will be validated annually.

8.14.3.5. Tower personnel will attempt to schedule training in advance with the OSW Airfield Services NCOIC. No-appointment training will only be conducted when manning and meteorological conditions permit. At the completion of this training, OSW personnel will issue Tower controllers a weather familiarity exam. If the controller scores 80% or higher, they will be certified to take tower visibility observations and the training will be documented on AF Form 3622, *Air Traffic Control/Weather Certification and Rating Record* (LRA), prescribed by AFI 11-218, *Aircraft Operations and Movement on the Ground* (located in the individual's AF Form 623, *On-The-Job Training Record*, prescribed by AFI 36-2201V3, *Air Force Training Program On the Job Training Administration*). A qualified weather technician will conduct the initial training in the weather duty section during controller in-processing.

8.14.3.6. RAPCON will verbally notify 48 OSS/OSW of significant observed radar weather echoes, at the earliest moment allowed by duty priorities.

8.14.3.7. 48 OSS/OSW will coordinate with the Tower and RAPCON Watch Supervisor before deactivating wind sensors for either the active or inactive runway.

8.14.3.8. AMOPS will verbally notify 48 OSS/OSW of the following:

8.14.3.8.1. Changes in Runway Surface Condition (RSC).

8.14.3.8.2. Changes in Runway Condition Reading (RCR).

8.14.3.8.3. Notice to Airmen (NOTAM), which affect airfield minimums.

8.14.3.8.4. Scheduled flights with Distinguished Visitors on board.

8.14.3.8.5. Any changes to published airfield operating hours (i.e. early/late opening/closing or out of normal operations).

8.14.3.9. 48 OSS/OSW will notify Airfield Management when temperature is forecast at or below 2 degrees Celsius IAW 48 FW Plan 32-1002.

8.14.3.10. Control tower personnel task certified to take visibility observations will also notify the weather technician when the observed tower prevailing visibility decreases to less than 4 SM (6000 m) or increases to or exceeds 4 SM (6000 m) and/or when it differs from the prevailing visibility by at least one reportable value.

8.14.4. 48 OSS/OSA Radar and Weather Systems Technicians (48 OSS/RAWS) will:

8.14.4.1. Place priority of maintenance repairs to AN/FMQ-19 Automated Meteorological Observing System.

8.14.4.2. Upon request from 48 OSS/OSW leadership, arrange for designated weather personnel to view all airfield weather sensors. All 48 OSS/OSW personnel must view airfield sensors once per tour IAW DAFMAN 15-129.

8.14.4.3. Conduct preventive maintenance and prompt repair of weather equipment, associated circuits, and ancillary equipment located at RAF Lakenheath, and maintained by RAWS.

8.14.4.4. Preventive maintenance or modifications that require equipment shutdown will be coordinated with 48 OSS/OSW flight leadership beforehand.

8.14.4.5. Notify 48 OSS/OSW in writing of any equipment outages so that Flight Leadership can take any necessary actions including notifying Squadron and Wing leaders.

JACK R. ARTHAUD,
Brigadier General, USAF
Commander, 48th Fighter Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFMAN 13-201, *Airspace Management,* 10 December 2020

DAFMAN 13-204V3*, Air Traffic Control,* 26 April 2024

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AFI 33-322, *Records Management and Information Governance Program,* 23 March 2020

DAFPD 15-1, *Weather Operations,* 28 May 2024

JO 7110.65AA, *Air Traffic Control,* 20 April 2023

Adopted Forms

DD Form 175-1, *Flight Weather Briefing*

AF Form 623, *On-the-Job Training Record*.

AF Form 847, *Recommendation for Change of Publication*

AF Form 3622, *Air Traffic Control/Weather Certification and Rating Record*

Abbreviations and Acronyms

AFI—Air Force Instruction

AFMAN—Air Force Manual

DAFMAN—Department of the Air Force Manual

DAFPD—Department of the Air Force Policy Directive

AFRIMS—Air Force Records Information Management System

AFW-WEBS—Air Force Weather Web Service

AMOPS—Airfield Management Operations

AMOS—Automated Meteorological Observation System

AOL—Alternate Operating Location

ASF—Airfield Services Function

ATC—Air Traffic Control

CDM—Chemical Downwind Message

CES/CEX—Civil Engineer Installation Office of Emergency Management

CES/DIO—Civil Engineering Squadron Defense Infrastructure Organization

CP—Command Post

CS—Communication Squadron

CS/SCXK—Base Freedom of Information Act Monitor

CWW—Cooperative Weather Watch

DCO—Delivery Control Officer

DLT—Desired Lead Time

DoD—Department of Defense

DoDDS—Department of Defense Dependent Schools

EDM—Effective Downwind Message

EO—Electro Optics

EOTDA—Electro-Optical Tactical Decision Aid

FLIP—Flight Information Publication

FS/CC—Fighter Squadron Commander

FW—Fighter Wing

FW/CC—Fighter Wing Commander

FW/PA—Fighter Wing Public Affairs

FW/SE—Fighter Wing Safety

FW/XP—Fighter Wing Plans and Inspections

IAW—In Accordance With

ICC—Installation Control Center

IDRC—Installation Deployment Readiness Cell

IRC—Instrument Refresher Course

JET—Joint Environmental Toolkit

LAN—Local Area Network

LLWS—Low Level Wind Shear

LRS/LGRDX—Fighter Wing Installation Deployment Officer

MAJCOM—Major Command

METAR—Aviation Routine Weather Report

METWATCH—Meteorological Watch

MIF—Mission Integration Function

MSG/CC—Mission Support Group Commander

MWP—Mission Weather Product

NAMO—NCOIC, Airfield Management Operations

NATCT—NCOIC, Air Traffic Control Training

NIPRNET—Nonsecure Internet Protocol Router Network

NOTAM—Notice to Airmen

NVG—Night Vision Goggles

OG—Operations Group

OG/CC—Operations Group Commander

OG/OGV—Operations Group Standardization and Evaluation

OJT—On-the-Job Training

OPR—Office of Primary Responsibility

OSS/CC—Operations Support Squadron Commander

OSS/OSA—Operations Support Squadron Airfield Operations

OSS/OSW—Operations Support Squadron Weather Flight

OWS—Operational Weather Squadron

PA—Public Affair

PIREP—Pilot Report

PMSV—Pilot-to-Metro Service

RAF—Royal Air Force

RAPCON—Radar Approach Control

RCR—Runway Condition Reading

RM—Risk Management

RSC—Runway Surface Condition

RVR—Runway Visual Range

SEC—Strike Eagle Complex

SIPRNET—Secret Internet Protocol Router Network

SOF—Supervisor of Flying

SOP—Standard Operating Procedure

SPECI—Aviation Special Weather Report

SWAP—Severe Weather Action Plan

SWAT—Severe Weather Action Team

TAF—Terminal Aerodrome Forecast

TAWS—Target Acquisition Weapons Software

TDA—Tactical Decision Aid

TO—Technical Order

UK—United Kingdom

USAF—United States Air Force

USAFE—United States Air Forces in Europe

UAO—Unmanned Airfield Operations

VAAC—Volcanic Ash Advisory Center

WA—Weather Advisory

WWATCH—Weather Watch

WW—Weather Warning

WWA—Watches, Warnings, and Advisories

Terms

Blizzard Conditions—Duration of ≥ 3 hours, sustained winds or gusts ≥ 30 knots, considerable falling and/or blowing snow, with prevailing visibility frequently ≤ ¼ mile/0400 meters.

Desired Lead-time—The amount of advanced notice a supported agency desires before the onset of a particular weather phenomenon.

Issue Time—The time when an agency is notified of a WA, WWATCH, or WW. When more than one agency is notified, the issue time is the time the last agency is notified. Follow up notifications are not considered when determining issue time. For RAF Lakenheath, the issue time is the time the WA, WWATCH, or WW was transmitted on JET.

Limited Duty Station—A weather station that provides less than 24-hour a day forecast service.

Meteorological Watch—Monitoring the weather for a route, area, or terminal and advising concerned organizations when hazardous conditions that could affect their operations or pose a hazard to life or property are observed or forecast to occur.

Mission Weather Product—A MWP is the integration of strategic center (557 WW) products, OWS produced products, and perishable weather data to the supported operator's/war-fighter's weapon system and tactics.

MISSIONWATCH—Monitoring the weather for a route, sortie, or training area and advising concerned organizations when forecast conditions change past operational thresholds or hazardous weather conditions that could affect operations or pose a threat to life or property are observed or forecast to occur.

Attachment 2

EXAMPLE WEATHER PRODUCTS

Figure A2.1. Mission Execution Forecast (MEF) (Day 1).

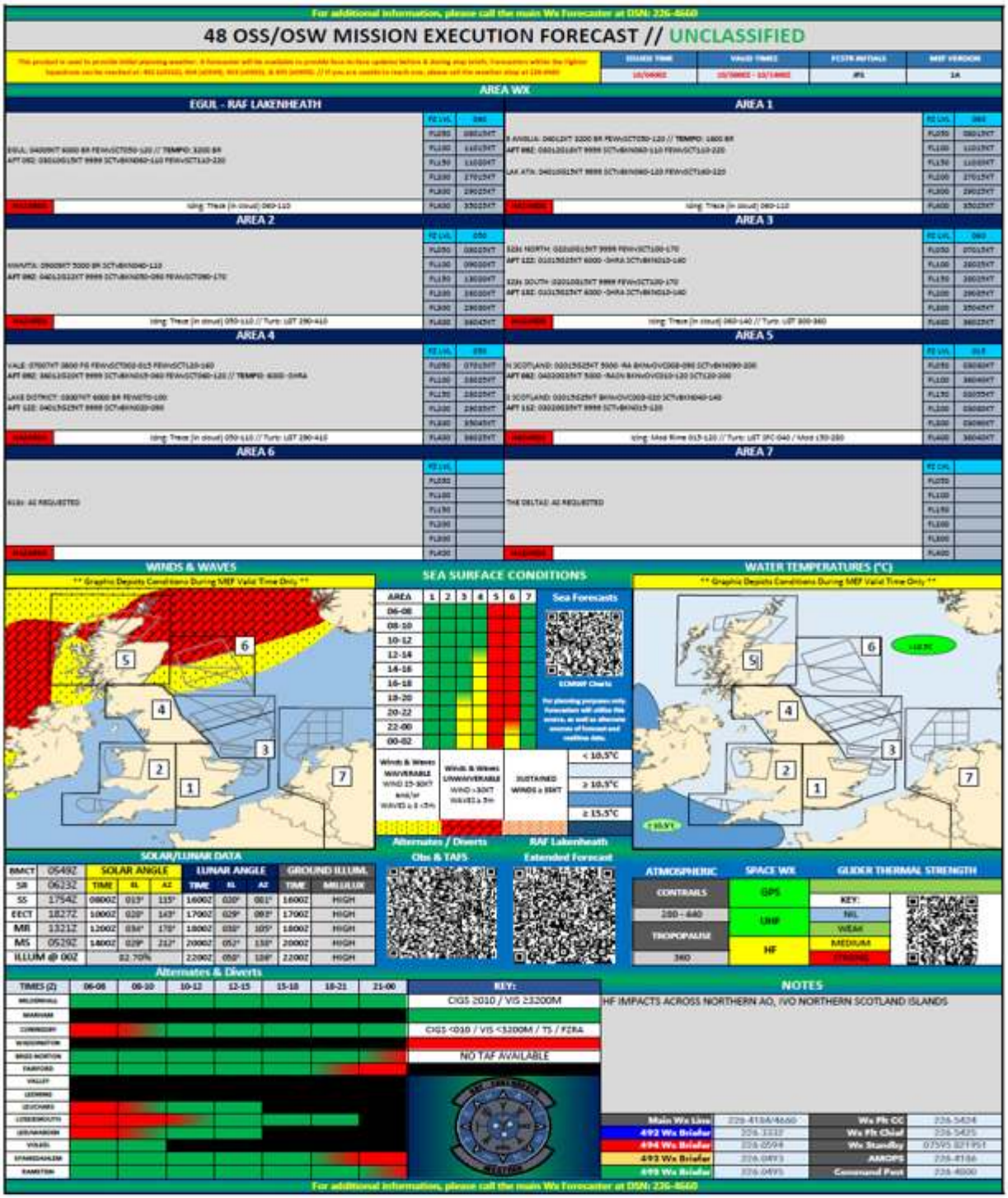


Figure A2.2. Planning Weather – Winds and Waves (Day 2).

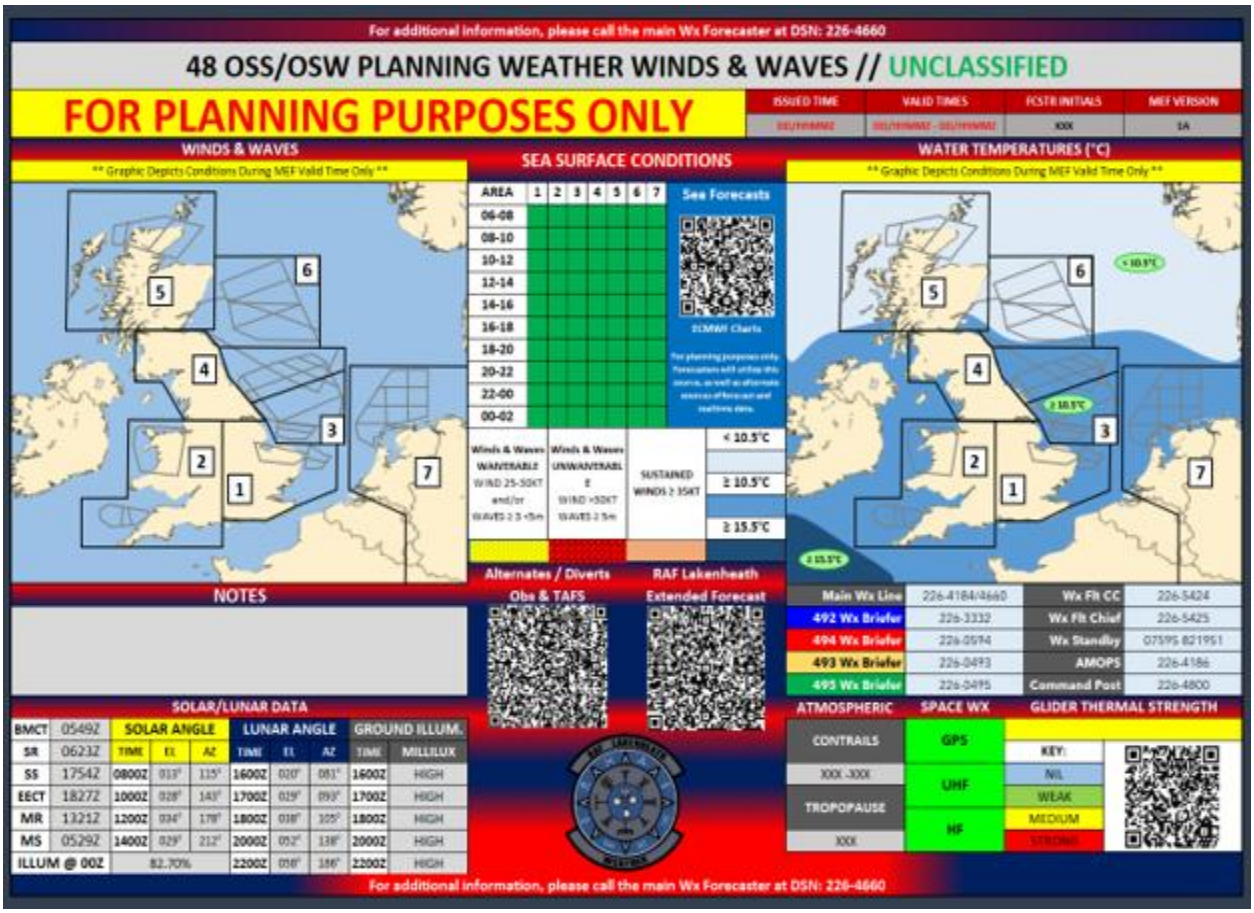


Figure A2.3. Planning Weather (Day 2).

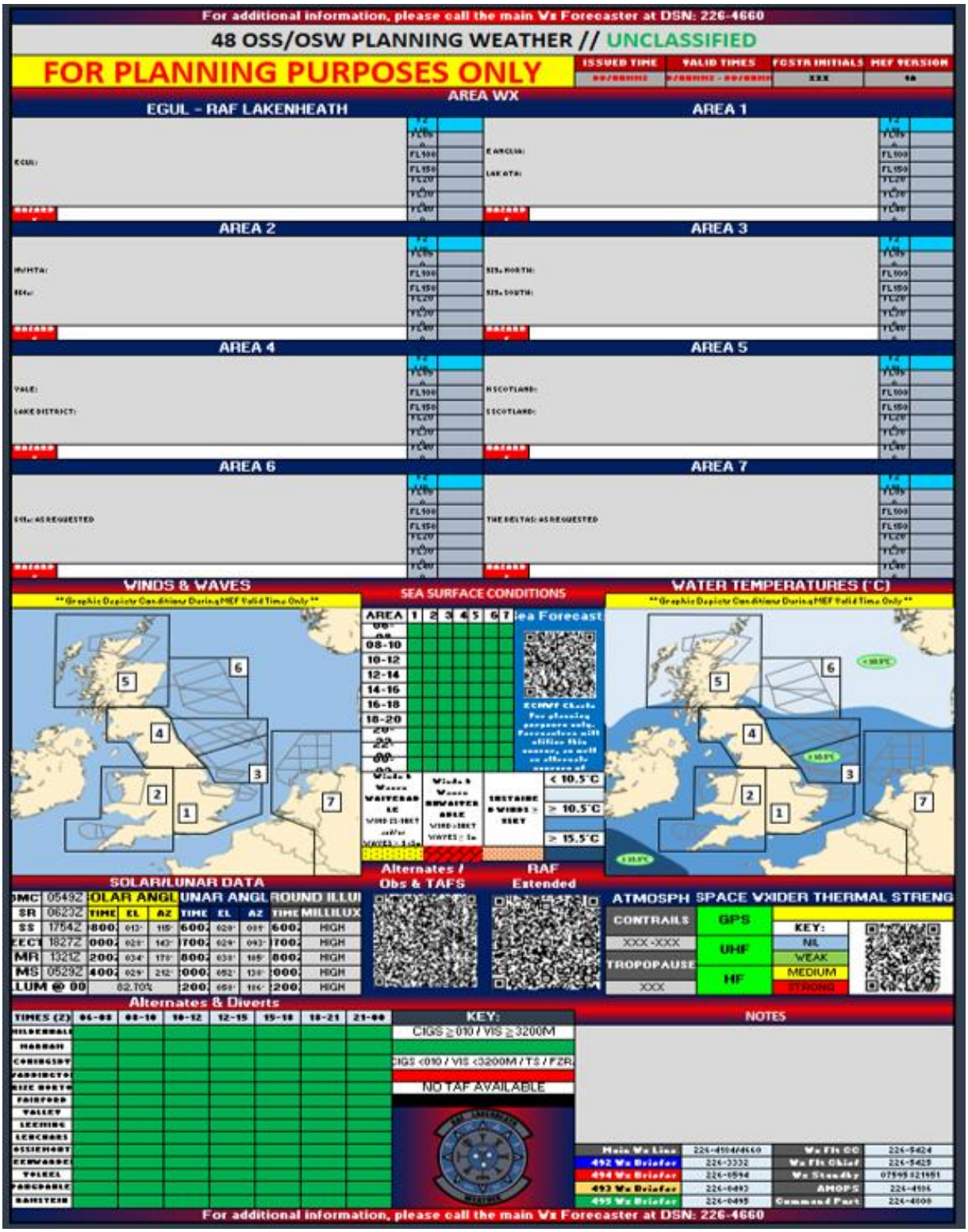


Figure A2.4. 72 Hour Divert Map.

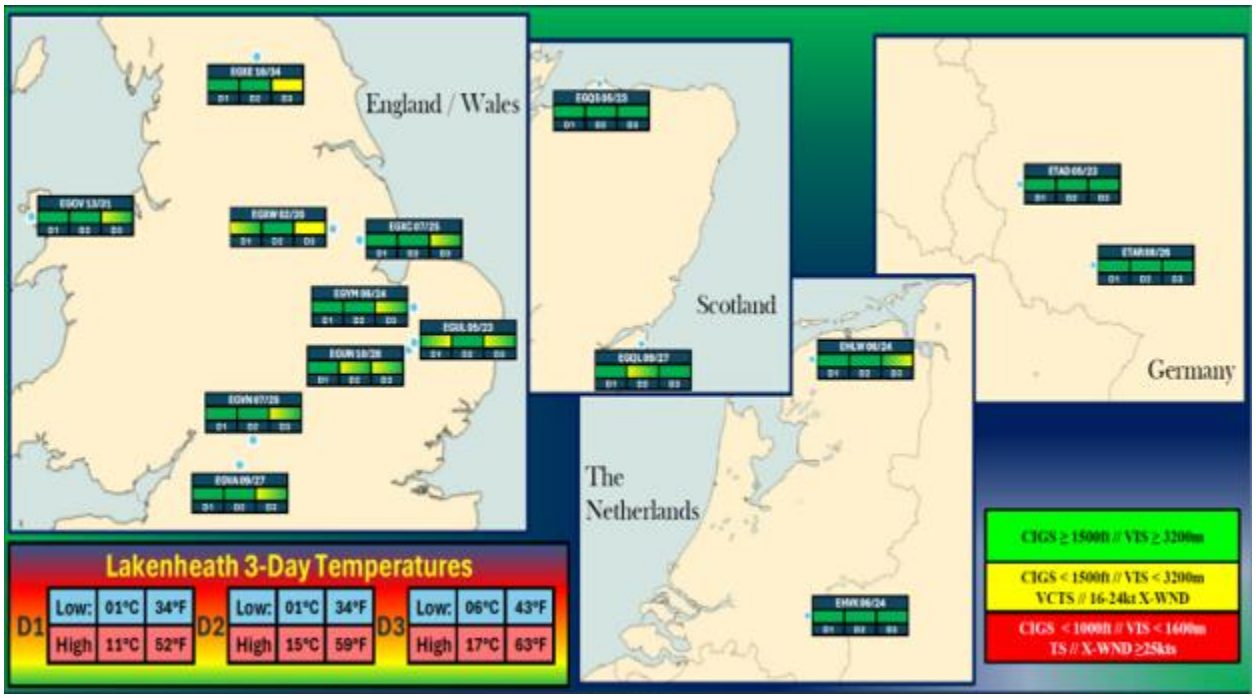


Figure A2.5. 5-Day – Operations Impacts.

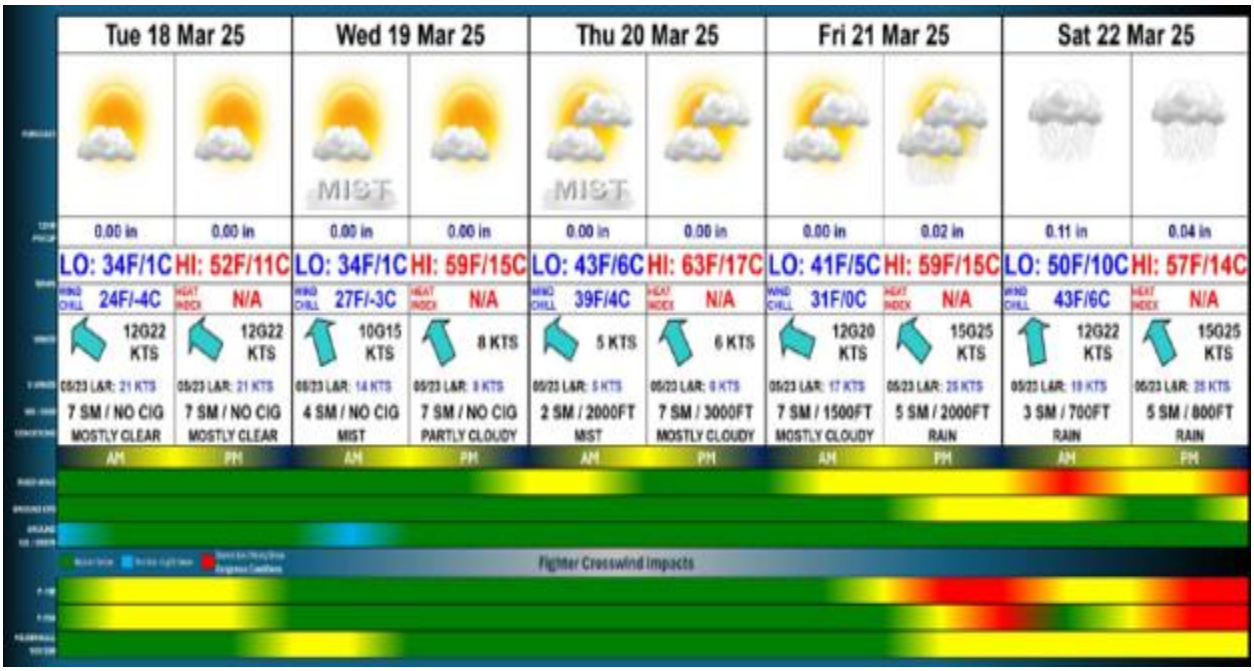
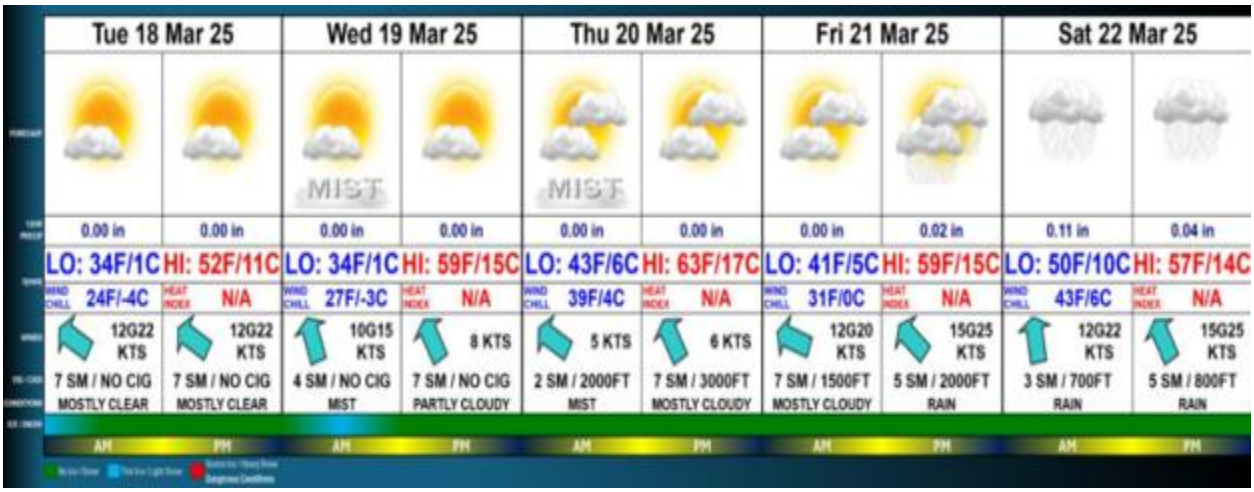


Figure A2.6. 5-Day – Public Release.



Attachment 3

WEATHER WATCH, WARNING AND ADVISORY

Figure A3.1. Weather Watch, Warning and Advisory Criteria.

Advisory Criteria			
Criteria	**Forecast/Observed**	**Desired Lead Time**	**Issued By**
Induction Icing Potential (1)	Observed	Observed	48 OSS/OSW*
Lightning within 10 NM	Forecast	30 minutes	48 OSS/OSW
Lightning within 10 NM	Observed	Observed	48 OSS/OSW
Lightning within 25 NM (2) (Aviation only)	Observed	Observed	48 OSS/OSW
SFC winds 25-34kts	Observed	Observed	48 OSS/OSW
Visibility ≤ 400 meters	Observed	Observed	48 OSS/OSW
Windshear Conditions below 2000ft AGL (Aviation Only)	Observed	Observed	48 OSS/OSW
AOL Activation	Observed	Observed	48 OSS/OSW
Fighter Index of Thermal Stress (FITS) Caution (Aviation Only)	Observed	Observed	48 OSS/OSW
Fighter Index of Thermal Stress (FITS) Danger (Aviation Only)	Observed	Observed	48 OSS/OSW
Note 1 *- Temp +10C ~ -20C with a Dew point Depression ≤ 3C*			
Note 2 *- Determine the altitude of +8C and -8C; if height values are separated by less than 10KFT (5KFT on either side of freezing level) then calculate heights on 5KFT above and below the freezing level; if values are separated by greater the 10KFT, use actual heights. When Aircraft-Induced LTG potential exists, include "Aircraft-Induced lightning risk high between Flight Levels XXX-YYY".*			
Watch Criteria			
Tornado	Forecast	As potential warrants	48 OSS/OSW
Severe Thunderstorm ≥ 50 knots and/or Hail ≥ ¾ inch in diameter**	Forecast	4 hours	48 OSS/OSW
Moderate Thunderstorm ≥ 35 knots but < 50 knots and/or Hail ≥ ¼ but < ¾ inch in diameter**	Forecast	4 hours	48 OSS/OSW
Nonconvective Damaging Winds ≥ 50 knots**	Forecast	4 hours	48 OSS/OSW
Snow Accumulation ≥ 2 inches in 12 hours***	Forecast	As potential warrants	48 OSS/OSW
Rain Accumulation ≥ 2 inches in 12 hours***	Forecast	As potential warrants	48 OSS/OSW
Freezing Precipitation	Forecast	As potential warrants	48 OSS/OSW
Lightning w/i 5 NM RAF of Lakenheath and Feltwel	Forecast	30 Minutes	48 OSS/OSW
Warning Criteria			
Tornado	Forecast	15 minutes	48 OSS/OSW
Severe Thunderstorm ≥ 50 knots and/or Hail ≥ ¾ inch in diameter**	Forecast	2 hours	48 OSS/OSW
Moderate Thunderstorm ≥ 35 knots but < 50 knots and/or Hail ≥ ¼ but < ¾ inch in diameter**	Forecast	1 hours	48 OSS/OSW
Nonconvective Damaging Winds ≥ 50 knots**	Forecast	2 hour	48 OSS/OSW
Nonconvective Strong Winds ≥ 35 knots but < 50 knots**	Forecast	1 hour	48 OSS/OSW
Snow Accumulation ≥ 2 inches in 12 hours***	Forecast	1 hour	48 OSS/OSW
Rain Accumulation ≥ 2 inches in 12 hours***	Forecast	1 hour	48 OSS/OSW
Freezing Precipitation	Forecast	1 hour	48 OSS/OSW
Lightning w/i 5 NM RAF Lakenheath	Observed	Observed	48 OSS/OSW*
Lightning w/i 5 NM RAF Feltwell	Observed	Observed	48 OSS/OSW*
Blizzard	Forecast	1 hour	48 OSS/OSW

Figure A3.2. Definitions/Thresholds for WWA Criteria.

Definitions/Thresholds for Criteria	
Tornado	Tornado / Funnel cloud (detected by radar or visually observed) AND threatening warned location / aerodrome complex
Severe Thunderstorm	Damaging Wind AF: GTE 50 kts Army: GTE 45 kts associated with thunderstorms -AND/OR- Damaging Hail
Moderate Thunderstorm	High Wind GTE 35 knots and less than severe criteria associated with thunderstorms -AND/OR- Large Hail
Damaging Winds	Surface winds not associated with thunderstorms meeting or exceeding damaging wind threshold.
Strong Winds	Surface Winds not associated with Thunderstorms GTE 35 knots (or locally determined threshold) and less than severe threshold
Freezing Precipitation	Threshold determined by local emergency management agencies Default parameter: Liquid precipitation of any type and intensity falls and produces glaze ice on exposed surfaces
Heavy Snow	Threshold determined by local Emergency management agencies New snowfall of a measurable amount (as determined by emergency management agencies) during a specified time period If no local threshold exists use 2ll in 12 hours
Blizzard	All of the following must occur: a) Surface visibility less than or equal to ¼ mile/400 meters b) Considerable falling and/or blowing snow c) Sustained wind speeds or gusts >30 knots
Heavy Rain	Threshold determined by local emergency management agencies Default parameter: New liquid precipitation accumulation of a measurable amount (as determined by emergency management agencies) during a specified time period If no local threshold exists use 2ll in 12 hours
Sandstorm	Winds carrying sand particles from the surface to no more than fifty feet above the surface, prevailing visibility < 5/8 to 5/16 statute miles (1000 to 500 meters). For prevailing visibility of □ 5/16 statute miles (500 meters), the storm is considered a heavy
Duststorm	Winds carrying dust particles from the surface to approximately ten thousand feet above
Lightning	Lightning within 5 nautical miles

Figure A3.3. Weather Watch/Warning/Advisory Dissemination Matrix.

